ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** -1	Rewrite as $y = -x + \ldots$; slope $= -\frac{a}{b}$.
2	**(a)** 14	Equal slopes between consecutive points.
3	**(a)** 8	Equal slopes: $-\frac{4}{2} = -\frac{k}{4}$, so $k = 8$.
4	**(b)** 3	Intercepts 2 and 3; area $= \frac{1}{2} \times 2 \times 3$.
5	**(b)** 259	Rate $= \dfrac{110}{10} = 11$ per km; fixed charge 28.
6	**(b)** 6	$2(-4) + (6)k = 28$ gives $k = 6$.
7	**(a)** $(-1, 7)$	$8(-1) + 5(7) = 27$
8	**(b)** 7	$y = mx + c$ with $y = 7$ at $x = 0$: $c = 7 - (6)(0)$.
9	**(d)** $-\frac{1}{2}$	Rewrite as $y = -\frac{1}{2}x + \ldots$; slope $= -\frac{a}{b}$.
10	**(c)** -7	Equal slopes between consecutive points.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rate $= \dfrac{163 - 82}{12 - 3} = 9$ per km; Fixed charge $= 82 - 9 \times 3 = 55$ (1); Fare for 23 km $= 9 \times 23 + 55$; $=$ ₹262 (1).	1+1
12	Slope of the first line $= -\frac{3}{5}$; Slope of the second $= -\frac{k}{10}$; equal slopes give $k = 6$ (1); Constants: $\frac{9}{11} \neq \frac{3}{6}$, i.e. the equations are not multiples of each other, so the lines are distinct (1).	1+1
13	$-2y = -6x + 18$; $y = 3x - 9$ (1); Slope $= 3$, y-intercept $= -9$ (1).	1+1
14	Rate $= \dfrac{259 - 140}{13 - 6} = 17$ per km; Fixed charge $= 140 - 17 \times 6 = 38$ (1); Fare for 28 km $= 17 \times 28 + 38$; $=$ ₹514 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	x-intercept: put $y = 0$: $8x = 32$, so $x = 4$ (1); y-intercept: put $x = 0$: $4y = 32$, so $y = 8$ (1); Area $= \frac{1}{2} \times 4 \times 8 = 16$ square units (1).	1+1+1
16	Slope of the first two points $= \dfrac{2 - (5)}{-2 - (-3)} = -3$ (1); Slope between $(-2, 2)$ and $(-1, k)$ must also be -3: $\dfrac{k - (2)}{1} = -3$ (1); $k = 2 - 3 \times 1 = -1$ (1).	1+1+1
17	Slope $= \dfrac{-25 - (-5)}{-6 - (-1)} = 4$; Use $(-1, -5)$: $-5 = 4(-1) + c$ (1); $c = -1$ (1); $y = 4x - 1$ (1).	1+1+1
	OR: x-intercept: put $y = 0$: $2x = 18$, so $x = 9$ (1); y-intercept: put $x = 0$: $9y = 18$, so $y = 2$ (1); Area $= \frac{1}{2} \times 9 \times 2 = 9$ square units (1).	1+1+1
18	Slope of the first two points $= \dfrac{-4 - (0)}{-7 - (-9)} = -2$ (1); Slope between $(-7, -4)$ and $(-5, k)$ must also be -2: $\dfrac{k - (-4)}{2} = -2$ (1); $k = -4 - 2 \times 2 = -8$ (1).	1+1+1
	OR: Slope $= \dfrac{-7 - (2)}{5 - (-4)} = -1$; Use $(-4, 2)$: $2 = -(-4) + c$ (1); $c = -2$ (1); $y = -x - 2$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Slope $= \dfrac{40}{20} = 2$ (1); Intercept $= 50 - 2 \times 20 = 10$, so $y = 2x + 10$ (1); $x = 128$: $y = 266$ (1); $286 = 2x + 10$ gives $x = 138$ (2).	1+1+1+2
	OR: $y = mx + c$ with $m = 5$: $25 = 5(4) + c$ gives $c = 5$ (1); Equation: $y = 5x + 5$ (1); x-intercept: $0 = 5x + 5$ gives $x = -1$ (1); At $x = 2$: $y = 15$ (2).	1+1+1+2
20	Slope $= \dfrac{531}{59} = 9$ (1); Intercept $= 107 - 9 \times 10 = 17$, so $y = 9x + 17$ (1); $x = 167$: $y = 1520$ (1); $1610 = 9x + 17$ gives $x = 177$ (2).	1+1+1+2

	OR: $y = mx + c$ with $m = 3$: $21 = 3(5) + c$ gives $c = 6$ (1); Equation: $y = 3x + 6$ (1); x-intercept: $0 = 3x + 6$ gives $x = -2$ (1); At $x = 5$: $y = 21$ (2).	1+1+1+2

Mistakes to avoid:

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
- Saying a linear equation in two variables has one solution; it has infinitely many.
- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as (y, x) instead of (x, y).
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 25	Equal slopes: $-\frac{5}{7} = -\frac{k}{35}$, so $k = 25$.
2	**(d)** $\frac{21}{2}$	Intercepts 7 and 3; area $= \frac{1}{2} \times 7 \times 3$.
3	**(d)** 794	Rate $= \dfrac{140}{7} = 20$ per km; fixed charge 34.
4	**(c)** -2	$6(6) + (-6)k = 48$ gives $k = -2$.
5	**(d)** $(6, 7)$	$5(6) + 8(7) = 86$
6	**(c)** -5	$y = mx + c$ with $y = -9$ at $x = -1$: $c = -9 - (4)(-1)$.
7	**(c)** $\frac{5}{2}$	Rewrite as $y = \frac{5}{2}x + \ldots$; slope $= -\frac{a}{b}$.
8	**(b)** -17	Equal slopes between consecutive points.
9	**(b)** 6	Equal slopes: $-\frac{3}{7} = -\frac{k}{14}$, so $k = 6$.
10	**(d)** 6	Intercepts 2 and 6; area $= \frac{1}{2} \times 2 \times 6$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Slope of the first line $= -\frac{5}{4}$; Slope of the second $= -\frac{k}{12}$; equal slopes give $k = 15$ (1); Constants: $\frac{9}{11} \neq \frac{5}{15}$, i.e. the equations are not multiples of each other, so the lines are distinct (1).	1+1
12	$3y = -2x + 46$; $y = -\frac{2}{3}x + \frac{46}{3}$ (1); Slope $= -\frac{2}{3}$, y-intercept $= \frac{46}{3}$ (1).	1+1
13	Rate $= \dfrac{190 - 92}{10 - 3} = 14$ per km; Fixed charge $= 92 - 14 \times 3 = 50$ (1); Fare for 36 km $= 14 \times 36 + 50$; $=$ ₹554 (1).	1+1
14	Slope of the first line $= -\frac{6}{3}$; Slope of the second $= -\frac{k}{12}$; equal slopes give $k = 24$ (1); Constants: $\frac{9}{11} \neq \frac{6}{24}$, i.e. the equations are not multiples of each other, so the lines are distinct (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	x-intercept: put $y = 0$: $4x = 20$, so $x = 5$ (1); y-intercept: put $x = 0$: $5y = 20$, so $y = 4$ (1); Area $= \frac{1}{2} \times 5 \times 4 = 10$ square units (1).	1+1+1
16	Slope of the first two points $= \dfrac{7 - (3)}{-1 - (-3)} = 2$ (1); Slope between $(-1, 7)$ and $(1, k)$ must also be 2: $\dfrac{k - (7)}{2} = 2$ (1); $k = 7 + 2 \times 2 = 11$ (1).	1+1+1
17	Slope $= \dfrac{-12 - (9)}{2 - (-5)} = -3$; Use $(-5, 9)$: $9 = -3(-5) + c$ (1); $c = -6$ (1); $y = -3x - 6$ (1).	1+1+1
	OR: x-intercept: put $y = 0$: $7x = 14$, so $x = 2$ (1); y-intercept: put $x = 0$: $2y = 14$, so $y = 7$ (1); Area $= \frac{1}{2} \times 2 \times 7 = 7$ square units (1).	1+1+1
18	Slope of the first two points $= \dfrac{-6 - (-4)}{-7 - (-8)} = -2$ (1); Slope between $(-7, -6)$ and $(-6, k)$ must also be -2: $\dfrac{k - (-6)}{1} = -2$ (1); $k = -6 - 2 \times 1 = -8$ (1).	1+1+1
	OR: Slope $= \dfrac{-9 - (-18)}{3 - (6)} = -3$; Use $(6, -18)$: $-18 = -3(6) + c$ (1); $c = 0$ (1); $y = -3x + 0$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Slope $= \dfrac{486}{54} = 9$ (1); Intercept $= 246 - 9 \times 26 = 12$, so $y = 9x + 12$ (1); $x = 128$: $y = 1164$ (1); $1254 = 9x + 12$ gives $x = 138$ (2).	1+1+1+2
	OR: $y = mx + c$ with $m = -3$: $-24 = -3(4) + c$ gives $c = -12$ (1); Equation: $y = -3x - 12$ (1); x-intercept: $0 = -3x - 12$ gives $x = -4$ (1); At $x = 4$: $y = -24$ (2).	1+1+1+2

20	Slope $= \frac{92}{46} = 2$ (1); Intercept $= 45 - 2 \times 14 = 17$, so $y = 2x + 17$ (1); $x = 125$: $y = 267$ (1); $287 = 2x + 17$ gives $x = 135$ (2).	1+1+1+2
	OR: $y = mx + c$ with $m = 4$: $24 = 4(3) + c$ gives $c = 12$ (1); Equation: $y = 4x + 12$ (1); x-intercept: $0 = 4x + 12$ gives $x = -3$ (1); At $x = 5$: $y = 32$ (2).	1+1+1+2

Mistakes to avoid:

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
- Saying a linear equation in two variables has one solution; it has infinitely many.
- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as (y, x) instead of (x, y).
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 614	Rate $= \frac{120}{6} = 20$ per km; fixed charge 34.
2	**(a)** 6	$2(2) + (2)k = 16$ gives $k = 6$.
3	**(b)** $(-4, 8)$	$2(-4) + 7(8) = 48$
4	**(c)** 27	$y = mx + c$ with $y = 9$ at $x = -3$: $c = 9 - (6)(-3)$.
5	**(d)** $-\frac{5}{7}$	Rewrite as $y = -\frac{5}{7}x + \ldots$; slope $= -\frac{a}{b}$.
6	**(a)** -18	Equal slopes between consecutive points.
7	**(b)** 18	Equal slopes: $-\frac{6}{2} = -\frac{k}{6}$, so $k = 18$.
8	**(b)** 12	Intercepts 6 and 4; area $= \frac{1}{2} \times 6 \times 4$.
9	**(b)** 612	Rate $= \frac{180}{10} = 18$ per km; fixed charge 36.
10	**(d)** -5	$3(6) + (-2)k = 28$ gives $k = -5$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$7y = -6x + 97$; $y = -\frac{6}{7}x + \frac{97}{7}$ (1); Slope $= -\frac{6}{7}$, y-intercept $= \frac{97}{7}$ (1).	1+1
12	Rate $= \frac{149 - 79}{11 - 4} = 10$ per km; Fixed charge $= 79 - 10 \times 4 = 39$ (1); Fare for 23 km $= 10 \times 23 + 39$; $=$ ₹269 (1).	1+1
13	Slope of the first line $= -\frac{1}{6}$; Slope of the second $= -\frac{k}{18}$; equal slopes give $k = 3$ (1); Constants: $\frac{9}{11} \neq \frac{1}{3}$, i.e. the equations are not multiples of each other, so the lines are distinct (1).	1+1
14	$3y = -5x + 77$; $y = -\frac{5}{3}x + \frac{77}{3}$ (1); Slope $= -\frac{5}{3}$, y-intercept $= \frac{77}{3}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	x-intercept: put $y = 0$: $2x = 14$, so $x = 7$ (1); y-intercept: put $x = 0$: $7y = 14$, so $y = 2$ (1); Area $= \frac{1}{2} \times 7 \times 2 = 7$ square units (1).	1+1+1
16	Slope of the first two points $= \frac{-15 - (-9)}{1 - (-2)} = -2$ (1); Slope between $(1, -15)$ and $(4, k)$ must also be -2: $\frac{k - (-15)}{3} = -2$ (1); $k = -15 - 2 \times 3 = -21$ (1).	1+1+1
17	Slope $= \frac{25 - (5)}{-6 - (-1)} = -4$; Use $(-1, 5)$: $5 = -4(-1) + c$ (1); $c = 1$ (1); $y = -4x + 1$ (1).	1+1+1
	OR: x-intercept: put $y = 0$: $9x = 36$, so $x = 4$ (1); y-intercept: put $x = 0$: $4y = 36$, so $y = 9$ (1); Area $= \frac{1}{2} \times 4 \times 9 = 18$ square units (1).	1+1+1
18	Slope of the first two points $= \frac{-10 - (-8)}{0 - (-1)} = -2$ (1); Slope between $(0, -10)$ and $(1, k)$ must also be -2: $\frac{k - (-10)}{1} = -2$ (1); $k = -10 - 2 \times 1 = -12$ (1).	1+1+1
	OR: Slope $= \frac{4 - (9)}{-1 - (4)} = 1$; Use $(4, 9)$: $9 = 1(4) + c$ (1); $c = 5$ (1); $y = x + 5$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Slope $= \frac{504}{56} = 9$ (1); Intercept $= 261 - 9 \times 26 = 27$, so $y = 9x + 27$ (1); $x = 193$: $y = 1764$ (1); $1854 = 9x + 27$ gives $x = 203$ (2).	1+1+1+2
	OR: $y = mx + c$ with $m = 5$: $20 = 5(1) + c$ gives $c = 15$ (1); Equation: $y = 5x + 15$ (1); x-intercept: $0 = 5x + 15$ gives $x = -3$ (1); At $x = 2$: $y = 25$ (2).	1+1+1+2

20	Slope $= \frac{413}{59} = 7$ (1); Intercept $= 186 - 7 \times 23 = 25$, so $y = 7x + 25$ (1); $x = 183$: $y = 1306$ (1); $1376 = 7x + 25$ gives $x = 193$ (2).	1+1+1+2
	OR: $y = mx + c$ with $m = -2$: $-14 = -2(3) + c$ gives $c = -8$ (1); Equation: $y = -2x - 8$ (1); x-intercept: $0 = -2x - 8$ gives $x = -4$ (1); At $x = 2$: $y = -12$ (2).	1+1+1+2

Mistakes to avoid:

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
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